A MINI-PROJECT REPORT

ON

"Vidya: Multilingual Financial News Sentiment Analysis System using FinBERT"

BY

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C E R T I F I C A T E

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This is to certify that

Aadesh Mahesh Angre (Roll No. 702)

Has satisfactorily completed this project entitled

"Vidya: Multilingual Financial News Sentiment Analysis System using FinBERT"

Towards the partial fulfilment of the

FOURTH YEAR BACHELOR OF ENGINEERING IN

(ARTIFICIAL INTELLIGENCE & DATA SCIENCE)

as laid by University of Mumbai

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Project Report Approval for B. E.

This project report entitled **"Vidya: Multilingual Financial News Sentiment Analysis System using FinBERT"** by **Aadesh Mahesh Angre -702** is approved for the degree of Fourth Year Bachelor of Engineering in Artificial Intelligence and Data Science.

Examiners:

1. _____

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Date:

Place:

Declaration

I declare that the work embodied in this project titled **"Vidya: Multilingual Financial News Sentiment Analysis System using FinBERT"** is my own contribution, carried out under the guidance of **Prof. Naresh Shende** at **Rajiv Gandhi Institute of Technology.**

I declare that this written submission represents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Indian investors and students follow company news in many languages, yet most tools that measure the mood of financial news work only in English and give only a score. A reader who does not know whether a headline is good or bad for a company, or why, has to read many newspapers and judge for himself. Company background, financial numbers and older news are kept on separate websites, which makes learning slow.

This project presents Vidya, a web application that reads financial news about 100 listed Indian companies and labels every headline as positive, neutral or negative with a confidence score. Vidya means knowledge and study, and the system is designed to teach the user why a headline is labelled the way it is. English headlines are classified by FinBERT, a BERT model fine-tuned on financial text. Headlines in other Indian languages are classified by a multilingual XLM-RoBERTa sentiment model. Each label is explained with plain-language reasons and the signal words found in the headline.

The system has a FastAPI (Python) back end with a SQLite database and a browser-based front end. A start page lets the user choose one of twelve Indian languages, and the font and the news edition follow the choice; the interface text is provided in English, Marathi and Hindi. After registration, login or guest entry, the user picks a company and sees its business story, history, headquarters, key financial ratios and a ranked comparison with world peers. News can be viewed for yesterday, today, this month, this year or for any start and end date. Interactive Chart.js graphs show share price against net news mood, daily mood counts, mood share and the newspapers that published the most headlines. Every newspaper card shows the publisher and a link to the original article. Pop-up help with a blurred background explains each section. Passwords are stored as salted PBKDF2-SHA256 hashes.

Keywords: Natural Language Processing, Sentiment Analysis, FinBERT, XLM-RoBERTa, Financial News, Multilingual, FastAPI, Chart.js, Explainable Labels, Indian Languages

Contents

Abstract
Contents
List of Figures
List of Tables
List of Algorithms
1 Introduction
1.1 Introduction Description
1.2 Organization of Report
2 Literature Review
2.1 Survey of Existing Systems
2.2 Limitations of Existing Systems / Research Gap
2.3 Problem Statement and Objectives
2.4 Scope
3 Proposed System
3.1 Analysis / Framework / Algorithm
3.2 Details of Hardware and Software
3.3 Design Details
3.3.1 System Flow / System Architecture
3.3.2 Detailed Design (UML)
3.4 Methodology / Procedure
3.5 Main Source Code
4 Implementation
4.1 Results
4.2 Discussion: Comparative Study / Analysis
5 Conclusion and Future Work
References
Acknowledgement

LIST OF FIGURES

Figure No.	Name
3.1	System Architecture
3.2	NLP Processing Pipeline
3.3	Implementation Process
3.4	Sequence Diagram: Viewing News Mood of a Company
4.1	Language Selection Page
4.2	Login and Register Page
4.3	Home Page with Sidebar and Company Story
4.4	Company Profile: Business, Finance and World Comparison
4.5	Timeline Selection, Summary Cards and Price vs News Mood Chart
4.6	Mood Charts and Top Newspapers
4.7	News Cards with Reasons for Each Label
4.8	Marathi Interface
4.9	News in Marathi
4.10	Guided Tour Pop-up with Blurred Background
4.11	Help Pop-up for the Price and Mood Chart

LIST OF TABLES

Table No.	Name
1.1	Comparison: Manual News Reading vs Vidya
2.1	Research Gap Analysis
3.1	Structure of a Headline Record
3.2	Fields of a Company Profile Record
3.3	REST API Endpoints
3.4	Software Requirements
4.1	Test Cases and Expected Results
4.2	Comparative Analysis

LIST OF ALGORITHMS

Sr. No.	Name
1	Fetching and Scoring Headlines
2	Trend Aggregation and Mood Calculation
3	Registration and Login

CHAPTER 1
Introduction

1.1 Introduction Description

A company's share price moves on information, and a large part of that information arrives as news. A good order, a profit announcement, a regulatory enquiry or a change of management can all change how investors feel about a company within hours. Newspapers publish hundreds of such headlines every day, and a student or a small investor cannot read all of them, compare them over months and still keep track of the numbers behind the company.

In India the problem has one more dimension, which is language. Business news is published in English, Hindi, Marathi, Tamil, Bengali and many other languages, and a large number of readers are more comfortable in their own language than in English. Most automatic sentiment tools, however, work only on English text and return only a score such as 0.82 positive. They do not tell the reader why the headline is positive, and they do not show the company's background, its financial strength or how it compares with companies abroad.

Natural Language Processing (NLP) can read a headline and decide whether the news is good, bad or neutral for the company. Modern transformer models such as BERT understand the meaning of a sentence from its context instead of counting words. FinBERT is a BERT model that is further trained on financial text, so it knows that words such as 'fell', 'probe' or 'record earnings' carry a particular meaning in finance. Multilingual models such as XLM-RoBERTa have been trained on text in about one hundred languages and can be used to classify sentiment in languages other than English.

This project applies these ideas in a working website named Vidya. The word Vidya (विद्या) is Sanskrit for knowledge, learning and study, and the website is built to help a user study the mood of the news before forming an opinion. It does not recommend what to buy or sell. For 100 listed Indian companies it collects headlines from Google News in the language chosen by the user, classifies each headline with a suitable model, explains the label in simple words, and shows the result as interactive graphs. For the same company it shows the business story, history, headquarters, key financial ratios and a ranked comparison with world peers, so that the user understands the company first and the news afterwards.

The application is made of a FastAPI back end, a SQLite database and an HTML, CSS and JavaScript front end that uses Chart.js for graphs. The user can register, log in or enter as a guest, can choose from twelve Indian languages on the first page, can look at news for

yesterday, today, this month, this year or any chosen dates, and can click small help buttons that open a pop-up with the background blurred to explain what each part does.

Table 1.1: Comparison: Manual News Reading vs Vidya

Aspect	Manual news reading	Vidya
Language	Reader must know the language of each paper	Twelve Indian languages for news; interface text in English, Marathi and Hindi
Time to see the mood	Hours of reading	Seconds, as counts and graphs
Reason for the mood	Reader's own judgement	Plain-language reason and signal words for each headline
Older news	Archive search in every paper	Any start and end date through Google News search
Company background	Separate websites	Story, finance details and world peers on the same page
Newspaper source	Known only by the paper in hand	Publisher name, logo and link on every card

1.2 Organization of Report

Chapter 1 introduces the background and motivation of the project. It explains why news mood matters, why language and explanations are important for Indian readers, and gives an overview of the developed system.

Chapter 2 reviews existing research on financial sentiment analysis and multilingual language models. It identifies the research gap, states the problem, and lists the objectives and scope of the project.

Chapter 3 describes the proposed system. It presents the framework and algorithms, the hardware and software requirements, the system architecture and UML design, the methodology followed and the main source code.

Chapter 4 presents the implementation results as screenshots of the application, together with test cases and a comparative discussion.

Chapter 5 concludes the work and outlines the future scope, such as a finance-tuned multilingual model, translation of the full interface into all supported languages and a tamper-proof multi-node ledger for storing results.

CHAPTER 2
Literature Review

2.1 Survey of Existing Systems

1. Tetlock, P. C. (2007), "Giving Content to Investor Sentiment: The Role of Media in the Stock Market". Approach: measured the pessimism of a widely read newspaper column by counting words from a general dictionary and showed that the media mood is related to movements in stock prices. Limitation: word counting ignores context, uses one newspaper, and works only on English text.

2. Loughran, T. and McDonald, B. (2011), "When Is a Liability Not a Liability?". Approach: showed that general-purpose word lists misclassify many words in financial documents and created finance-specific positive and negative word lists. Limitation: lists still cannot understand the meaning of a full sentence and exist only for English.

3. Malo, P., Sinha, A., Korhonen, P., Wallenius, J. and Takala, P. (2014), "Good Debt or Bad Debt: Detecting Semantic Orientations in Economic Texts". Approach: created the Financial PhraseBank, a set of financial news sentences labelled positive, negative or neutral by finance experts, which is widely used to train and test models. Limitation: English sentences only and a limited number of sentences.

4. Devlin, J., Chang, M., Lee, K. and Toutanova, K. (2019), "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". Approach: introduced a transformer model that reads a sentence in both directions and can be fine-tuned for many tasks, including sentiment analysis. Limitation: the original model is trained on general text and needs fine-tuning for finance.

5. Araci, D. (2019), "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models". Approach: further trained BERT on financial text and fine-tuned it on the Financial PhraseBank, improving finance-specific sentiment classification over general models. Limitation: English only, and the model gives a label and confidence but no explanation.

6. Conneau, A. et al. (2020), "Unsupervised Cross-lingual Representation Learning at Scale". Approach: presented XLM-RoBERTa, a transformer trained on text in about one hundred languages, including several Indian languages, so that one model can be reused across languages. Limitation: it is a general model and needs a task-specific classification layer.

7. Barbieri, F., Espinosa Anke, L. and Camacho-Collados, J. (2022), "XLM-T: Multilingual Language Models in Twitter for Sentiment Analysis and Beyond". Approach: fine-tuned XLM-RoBERTa for sentiment on tweets in several languages, including Hindi, and released ready-to-use models.

2.2 Limitations of Existing Systems / Research Gap

Table 2.1: Research Gap Analysis

System	Approach	Limitation / Research gap
Manual news reading	Investor reads several newspapers and judges the mood	Slow, hard to compare over months, and limited by the languages the reader knows
Lexicon-based tools	Count positive and negative words from a list	Ignore context and sentence meaning; mostly English
English-only FinBERT tools	Transformer fine-tuned on financial text	Do not read Indian-language news and give only a score without reasons
Multilingual social-media models	XLM-RoBERTa models for tweets	Not tuned for financial news and offer no company background
Proposed system (Vidya)	FinBERT plus a multilingual model, explanations, company story and date timeline in one web application	Non-English labels are less accurate than FinBERT on English, and coverage depends on Google News

2.3 Problem Statement and Objectives

Indian readers who want to understand how the news is treating a listed company have to read many newspapers in different languages, cannot easily see the trend over time, and get no reason behind a positive or negative judgement. Existing automatic tools are mostly limited to English and keep company background, financial numbers and news in separate places. There is a need for a single, easy-to-use website that reads financial news in the reader's own language, labels it with a language model, explains the label in simple words and shows the company's background and numbers next to the news.

2.3.1 Objectives

- To collect financial news headlines for 100 listed Indian companies from Google News for any chosen company, language and date range.
- To classify every headline as positive, neutral or negative using FinBERT for English and a multilingual XLM-RoBERTa model for other languages.
- To explain each label in plain language and to show the signal words that influenced it.
- To show the company's business, history, headquarters, financial ratios and a ranked comparison with world peers.
- To present the result as interactive graphs with a timeline of yesterday, today, month, year or any chosen dates, and to show the newspaper and link for every headline.
- To provide registration, login and guest access, a language-first start page, and pop-up help with a blurred background so that a beginner can use the site without training.

2.4 Scope

The project covers 100 large listed Indian companies grouped in nine sectors. News is read from Google News editions for twelve languages, the interface text is available in English, Marathi and Hindi, and the font follows the chosen language. The system is meant for learning and for understanding news sentiment. It does not give buying or selling advice, does not predict prices, and does not guarantee that every local newspaper is available, because it depends on what Google News indexes.

CHAPTER 3
Proposed System

3.1 Analysis / Framework / Algorithm

The proposed system works in the following stages. The overall working can be summarized as follows.

1. Language and login: the user first chooses a language, which sets the font and the news edition, and then registers, logs in or continues as a guest.
2. Company selection: the user searches the list of 100 companies in the sidebar and clicks one.
3. Company profile: the back end collects the business summary, financial ratios and headquarters from Yahoo Finance, the history and photo from Wikipedia, and the market values of sector peers worldwide. The result is stored for one day.
4. News collection: the back end builds a Google News RSS query for the company with the chosen language edition and date range, and reads up to 100 headlines.
5. Sentiment classification: FinBERT is used for English headlines and a multilingual XLM-RoBERTa sentiment model for other languages. Headlines are processed in batches of sixteen. If models cannot be loaded, a keyword method is used as a fallback.
6. Explanation: for each label, the words from a list of positive and negative finance words found in the headline are returned together with a plain-language reason.
7. Aggregation: headlines are counted by day or by month, the net mood (positive minus negative) is calculated, share prices are aligned with the same dates, and the newspapers with most headlines are found.
8. Display: the front end draws the graphs and news cards and shows pop-up help when a help button is clicked.

The main algorithms are listed below.

Algorithm 1: Fetching and Scoring Headlines

```
Input : company, language, start date a, end date b
Output: scored headlines saved in the database


1. key = (ticker, language, a, b)
2. if key was fetched in the last 30 minutes: stop
3. query = company + " after:" + a + " before:" + (b + 1 day)
4. select the Google News edition (hl, ceid) for the language
5. entries = first 100 items of the RSS feed for the query
6. model = FinBERT if language is English else XLM-R sentiment
7. results = model(titles of entries) in batches of 16
8. for each entry and result:
     save (ticker, language, title, date, label, score,
           newspaper, newspaper site, link)
9. duplicates (same ticker, language, title, day) are ignored
```

Algorithm 2: Trend Aggregation and Mood Calculation

```
Input : ticker, language, range (today, yesterday, month,
        year or custom start and end dates)
Output: counts, price series, top newspapers, mood

1.  (a, b) = start and end dates of the chosen range
2.  run Algorithm 1 for (ticker, language, a, b)
3.  rows = saved headlines of ticker and language between a and b
4.  bucket = month if (b - a) > 90 days else day
5.  for each bucket count positive, neutral and negative
6.  net mood of a bucket = positive - negative
7.  price = closing prices from Yahoo Finance for the same buckets
8.  papers = 8 newspapers with the most headlines
9.  signal words = 4 most common words for positive, for negative
10. if positive > 1.3 x negative: mood = Bullish
    else if negative > 1.3 x positive: mood = Bearish
    else mood = Mixed
11. return buckets, price, papers, mood, first 25 headlines
```

Algorithm 3: Registration and Login

```
Input : email, password, name (only for registration)
Output: session token or an error message

Register:
1. reject if email has no "@" or password is shorter than 6
2. reject if the email is already registered
3. salt = 16 random bytes
4. hash = PBKDF2-HMAC-SHA256(password, salt, 120000 rounds)
5. save (email, name, salt, hash)
6. token = 48 random hex characters; save (token, email)

Login:
7. read salt and hash saved for the email
8. recompute the hash from the entered password
9. if the hashes differ: return "email or password is wrong"
10. else create a token as in step 6 and return it
```

3.2 Details of Hardware and Software

3.2.1 Hardware Requirement

A personal computer or laptop with at least 8 GB of RAM, a modern multi-core processor and about 4 GB of free disk space for the downloaded language models is sufficient. A graphics card is not required. An internet connection is needed to read news, prices and company details. Other computers on the same Wi-Fi network can open the website by using the IP address of the computer that runs the server.

3.2.2 Software Requirement

Table 3.4: Software Requirements

Software	Purpose
Windows 10 or 11	Operating system used for development
Python 3.10 or higher	Back end programming language
Visual Studio Code	Code editor
FastAPI and Uvicorn	Web framework and server for the REST API
SQLite (sqlite3 module)	Database for headlines, users, sessions and company profiles
Transformers and PyTorch	Running FinBERT and the multilingual XLM-RoBERTa sentiment model
feedparser	Reading Google News RSS feeds
yfinance	Share price, financial ratios and peer market values
Wikipedia API	Company history and photo
HTML, CSS, JavaScript	Front end pages, styles and behaviour
Chart.js 4.4.1	Interactive graphs
Google Fonts (Noto Sans family)	Fonts for the Indian scripts
Web browser (Chrome, Edge or Firefox)	Viewing the website

3.3 Design Details

3.3.1 System Flow / System Architecture

The system has four layers. The presentation layer is the web page in the user's browser. The application layer is the FastAPI server, which offers REST endpoints. The NLP layer holds the language models and the explanation logic. The data layer is a SQLite database, and the server also talks to three external sources: Google News, Yahoo Finance and Wikipedia.

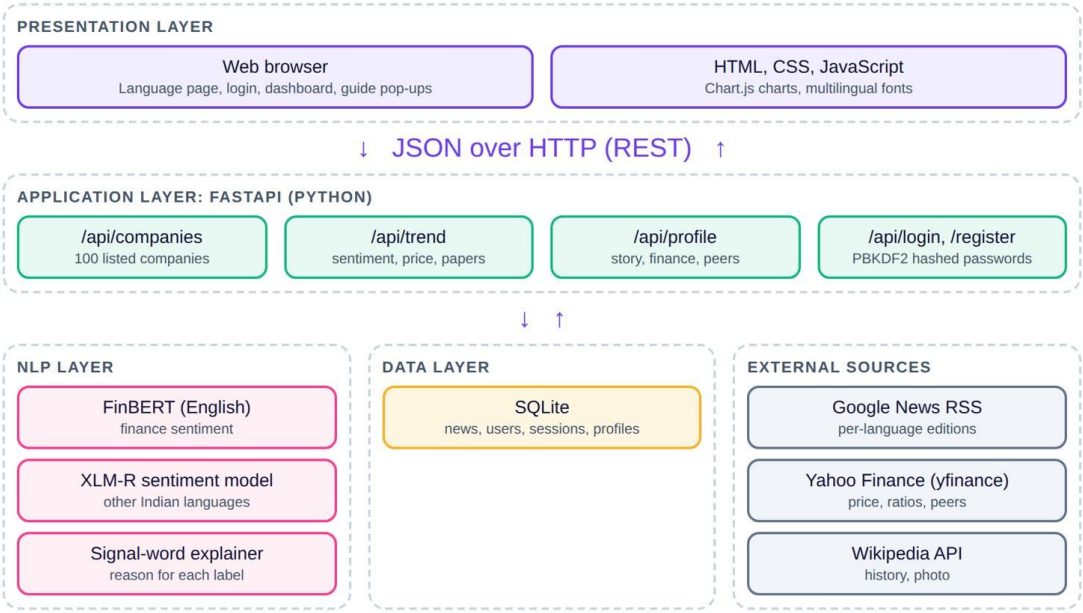


Figure 3.1: System Architecture

Figure 3.2 shows how one headline travels through the NLP pipeline, from the RSS feed to the news card shown to the user.



Figure 3.2: NLP Processing Pipeline

Figure 3.3 shows the steps followed to implement the project.

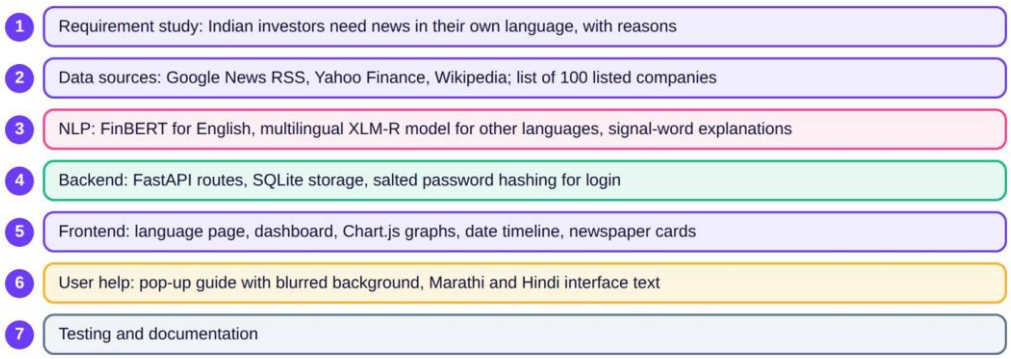


Figure 3.3: Implementation Process

Table 3.1: Structure of a Headline Record (table news)

Field	Description
ticker	Short code of the company, for example TCS
lang	Language code of the news edition, for example en, mr, hi
title	Headline text as published
day	Date of publication (YYYY-MM-DD)
label	positive, neutral or negative
score	Confidence of the model between 0 and 1
paper	Name of the newspaper or publisher
site	Website address of the publisher, used for its logo
link	Link to the original article

Table 3.2: Fields of a Company Profile Record

Field	Description
about	Business summary from Yahoo Finance
history and photo	Introduction text and picture from Wikipedia
where	Headquarters, employees, industry and website
stats	Market value, revenue, margins, P/E, EPS, debt to equity, return on equity, price range, growth, debt, cash, cash flow and analyst view
people	Names and titles of key officers when available
peers	Sector peers worldwide with market value in USD billions

Table 3.3: REST API Endpoints

Endpoint	Method	Purpose
/api/companies	GET	List of the 100 companies with sector
/api/trend/{ticker}	GET	Counts, price, papers, mood and headlines for range, lang, start and end
/api/profile/{ticker}	GET	Company story, finance details and world peers
/api/predict	POST	Label and confidence for one text
/api/register	POST	Create a user and return a token
/api/login	POST	Check the password and return a token
/api/me	GET	Check a saved token and return the user name

3.3.2 Detailed Design (UML)

The sequence diagram in Figure 3.4 shows what happens when a user chooses a company, a language and a date range. The front end calls the trend endpoint. The server checks the database, fetches new headlines from Google News if needed, scores them with the language model, saves them and returns the aggregated result.

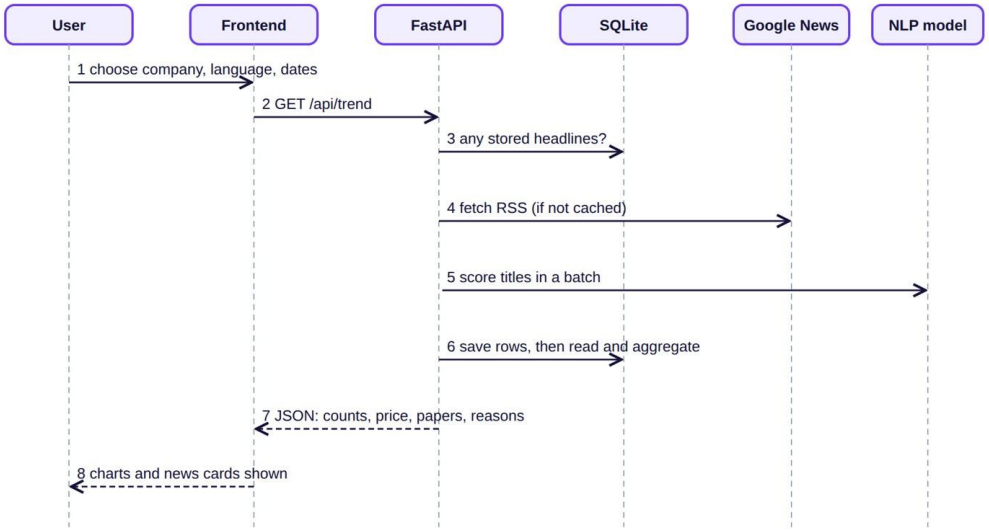


Figure 3.4: Sequence Diagram: Viewing News Mood of a Company

3.4 Methodology / Procedure

1. Study the problem of reading financial news in several Indian languages and decide the features of the website.
2. Prepare the list of 100 listed companies with ticker and sector.
3. Select data sources: Google News RSS for headlines, Yahoo Finance for prices and ratios, Wikipedia for history.
4. Select the models: FinBERT for English and a multilingual XLM-RoBERTa sentiment model for other languages, with a keyword fallback when models cannot be loaded.
5. Build the FastAPI back end with the SQLite database, the trend, profile and authentication endpoints.
6. Build the front end: language page, login page, sidebar, company page, graphs and news cards.
7. Add the explanation of each label, the date timeline and the pop-up guide with a blurred background.
8. Test the features with the cases listed in Chapter 4 and record the screenshots.

3.5 Main Source Code

The most important parts of the source code are shown below. The first listing selects the model by language and scores a list of headlines; if no model can be loaded, a keyword method is used.

```
def model(lang):
    k = "en" if lang == "en" else "xx"
    if k not in _m:
        _m[k] = None
```

```
        if not os.getenv("VIDYA_DEMO"):
            try:
                from transformers import pipeline
                _m[k] = pipeline("text-classification", model="ProsusAI/finbert" if k == "en"
else "cardiffnlp/twitter-xlm-roberta-base-sentiment")
            except Exception:
                pass
    return _m[k]
def score_many(texts, lang="en"):
    m = model(lang)
    if m:
        return [(r["label"].lower(), round(r["score"], 2)) for r in m([t[:512] for t in
texts], batch_size=16)]
    out = []
    for t in texts:
        t = t.lower(); p = sum(w in t for w in POS); n = sum(w in t for w in NEG)
        out.append((("positive" if p > n else "negative" if n > p else "neutral"), min(.95,
round(.6 + .1 * abs(p - n), 2))))
    return out
```

The next listing fetches headlines from Google News for the chosen company, language and dates, scores them in one batch and stores them.

```
def fetch(c, lang, a, b):
    key = (c[1], lang, a, b)
    if time.time() - last.get(key, 0) < 1800: return
    last[key] = time.time()
    hl, ceid = LANG.get(lang, LANG["en"])
    q = urllib.parse.quote(f"{c[0]} after:{a.isoformat()} before:{(b +
dt.timedelta(1)).isoformat()}")
    try:
        import feedparser
        es =
feedparser.parse(f"https://news.google.com/rss/search?q={q}&hl={hl}&gl=IN&ceid={ceid}").entries[:100]
        if not es: return
        for e, (l, s) in zip(es, score_many([e.title for e in es], lang)):
            src = e.get("source", {})
            db.execute("insert or ignore into news values(?,?,?,?,?,?,?,?,?)",
                       (c[1], lang, e.title, dt.date(*e.published_parsed[:3]).isoformat(), l, s,
src.get("title", "Google News"), src.get("href", ""), e.link))
        db.commit()
    except Exception:
        pass
```

The trend function reads the saved headlines, groups them by day or month, adds price, newspapers and signal words, and returns the mood.

```
@app.get("/api/trend/{ticker}")
def trend(ticker: str, range: str = "today", lang: str = "en", start: str = "", end: str =
""):
    c = find(ticker)
    if not c: return {"error": "Unknown company"}
    t = dt.date.today()
    if range == "custom":
        try: a, b = dt.date.fromisoformat(start), dt.date.fromisoformat(end)
        except ValueError: return {"error": "Pick a valid start and end date."}
        if a > b: a, b = b, a
    else:
        a, b = {"today": (t, t), "yesterday": (t - dt.timedelta(1),) * 2, "month": (t -
dt.timedelta(30), t)}.get(range, (t - dt.timedelta(365), t))
    fetch(c, lang, a, b)
    rows = db.execute("select title,day,label,score,paper,site,link from news where ticker=?
and lang=? and day between ? and ? order by day desc",
                      (ticker, lang, a.isoformat(), b.isoformat())).fetchall()
    n = 7 if (b - a).days > 90 else 10
    bk = {}
    for r in rows: bk.setdefault(r[1][:n], {"positive": 0, "negative": 0, "neutral":
0})[r[2]] += 1
    labels = sorted(bk)
    tot = {k: sum(bk[l][k] for l in labels) for k in ("positive", "negative", "neutral")}
    all_ = sum(tot.values()) or 1
```

```
    mood = "Bullish" if tot["positive"] > tot["negative"] * 1.3 else "Bearish" if
tot["negative"] > tot["positive"] * 1.3 else "Mixed"
    drv = {k: [w for w, _ in Counter(x for r in rows if r[2] == k for x in words(r[0],
k)).most_common(4)] for k in ("positive", "negative")}
    return {"labels": labels, **{k: [bk[l][k] for l in labels] for k in tot}, "total":
sum(tot.values()), "price": prices(c, a, b, n == 7, labels), "papers": [{"paper": p, "count":
k} for p, k in Counter(r[4] for r in rows).most_common(8)],
            "pos_pct": round(100 * tot["positive"] / all_), "neg_pct": round(100 *
tot["negative"] / all_), "mood": mood, "drivers": drv,
            "model": "FinBERT" if model(lang) and lang == "en" else "multilingual model" if
model(lang) else "demo keywords",
            "headlines": [dict(zip(("title", "day", "label", "score", "paper", "site",
"link"), r), words=words(r[0], r[2])) for r in rows[:25]]}
```

Passwords are hashed with a random salt using PBKDF2-HMAC-SHA256, and a session token is created at login.

```
hp = lambda pw, salt: hashlib.pbkdf2_hmac("sha256", pw.encode(), salt, 120000).hex()
def session(email, name):
    tok = secrets.token_hex(24)
    db.execute("insert into sessions values(?,?)", (tok, email)); db.commit()
    return {"token": tok, "name": name}
@app.post("/api/register")
def register(b: Cred):
    e = b.email.strip().lower()
    if "@" not in e or len(b.password) < 6: return {"error": "Enter a valid email and a
password of at least 6 characters."}
    if db.execute("select 1 from users where email=?", (e,)).fetchone(): return {"error":
"That email is already registered. Try logging in."}
    salt = secrets.token_bytes(16); name = b.name.strip() or e.split("@")[0]
    db.execute("insert into users values(?,?,?,?)", (e, name, salt.hex(), hp(b.password,
salt))); db.commit()
    return session(e, name)
@app.post("/api/login")
def login(b: Cred):
    e = b.email.strip().lower()
    u = db.execute("select name,salt,hash from users where email=?", (e,)).fetchone()
    if not u or hp(b.password, bytes.fromhex(u[1])) != u[2]: return {"error": "Email or
password is wrong. Check and try again."}
    return session(e, u[0])
```

The help pop-up in the front end highlights the target section, scrolls to it, and shows the explanation in the chosen language.

CHAPTER 4
Implementation

4.1 Results

The application was run on a local computer with the FastAPI server and opened in a web browser. The following figures show the main screens of the website.

The first page asks the user to choose a language. The animated Vidya logo and the name of the site in English, Marathi and Hindi appear at the top, and the purpose of the site is written at the bottom.

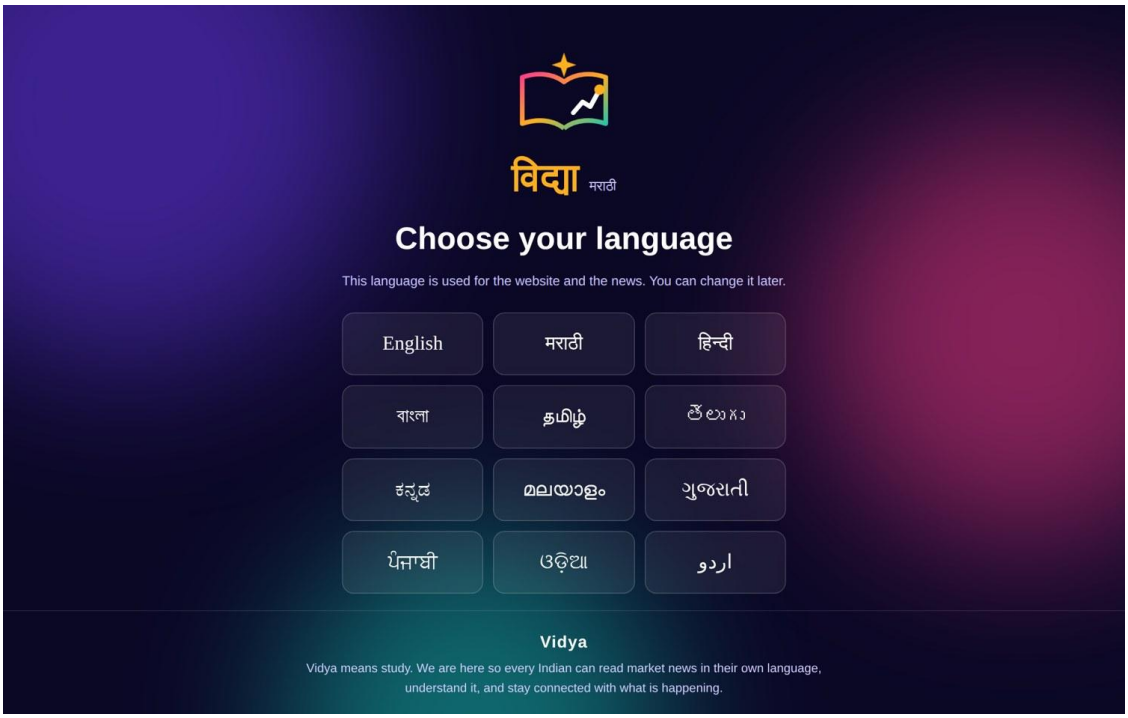


Figure 4.1: Language Selection Page

After the language is chosen, the login page opens. The user can log in, register a new account or continue as a guest.

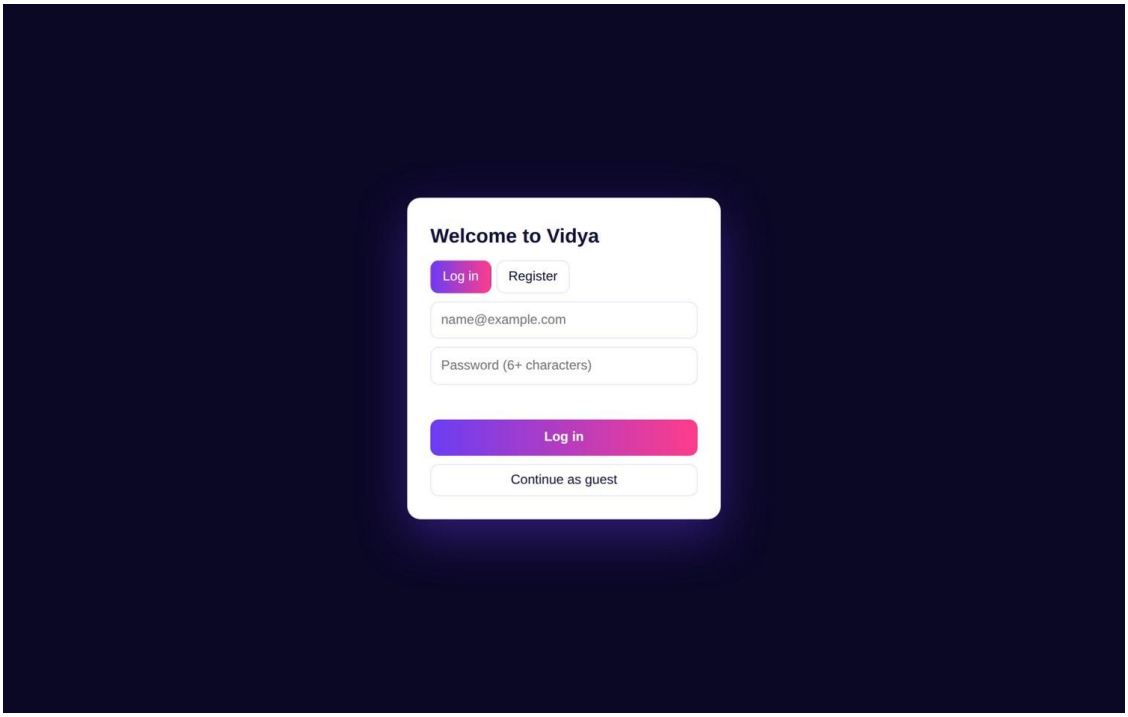


Figure 4.2: Login and Register Page

The home page has a sidebar with the user name, the language selector and the list of 100 companies. The Guide and Log out buttons are at the top, and the company story opens first.

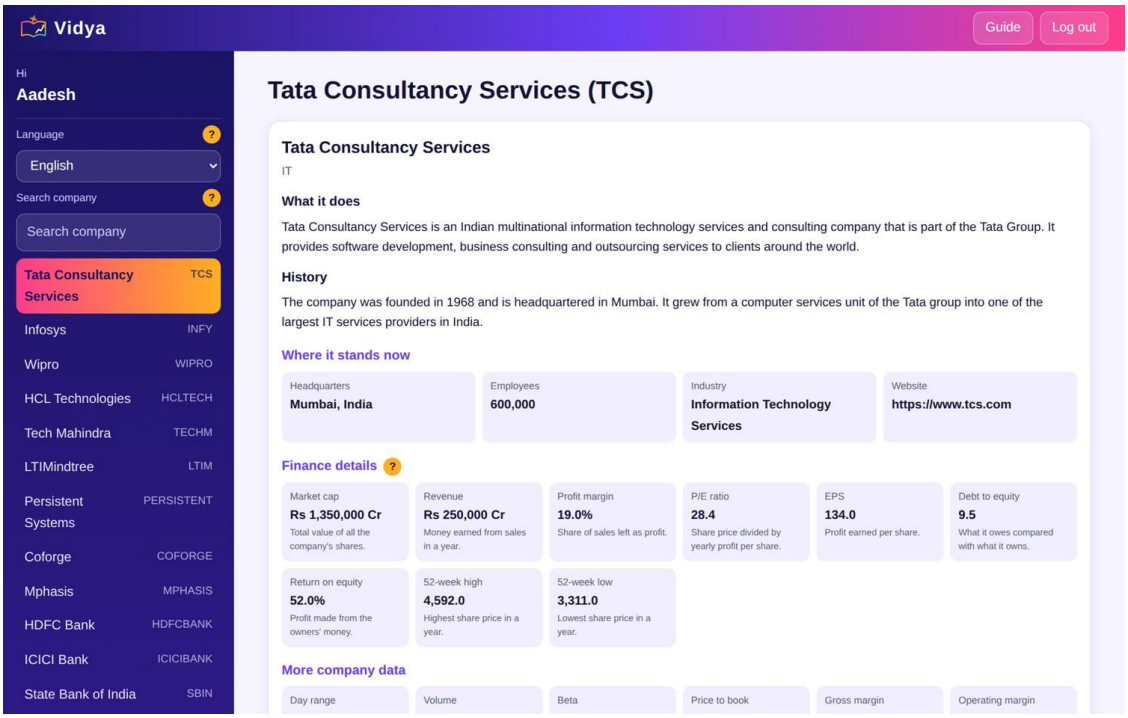


Figure 4.3: Home Page with Sidebar and Company Story

The company profile shows what the company does, its history, where it stands now, finance details with a one-line explanation under each number, more company data, and a ranked world comparison of sector peers from large to small.

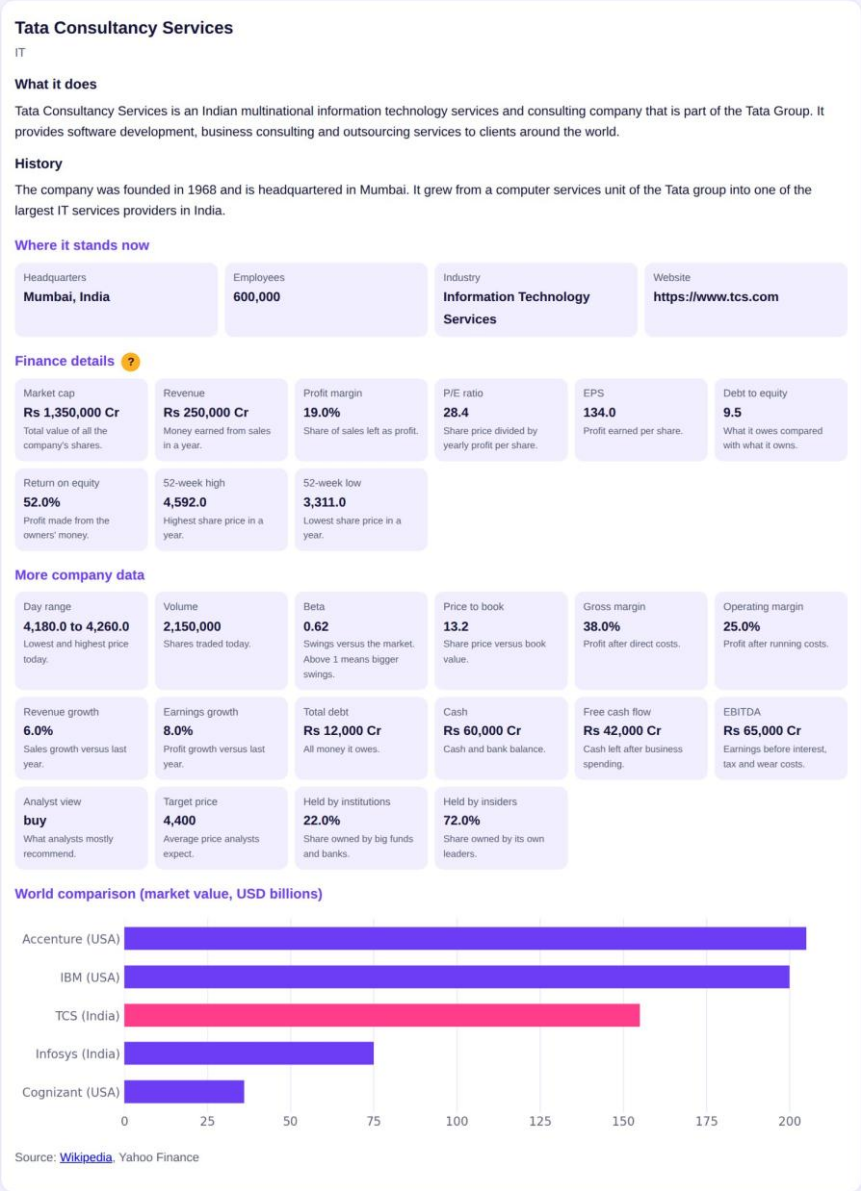


Figure 4.4: Company Profile: Business, Finance and World Comparison

The timeline buttons allow yesterday, today, this month, this year or any chosen start and end dates. The summary cards show the number of headlines and the share of positive and negative news. The combined chart shows the share price as a line and the net news mood as bars.

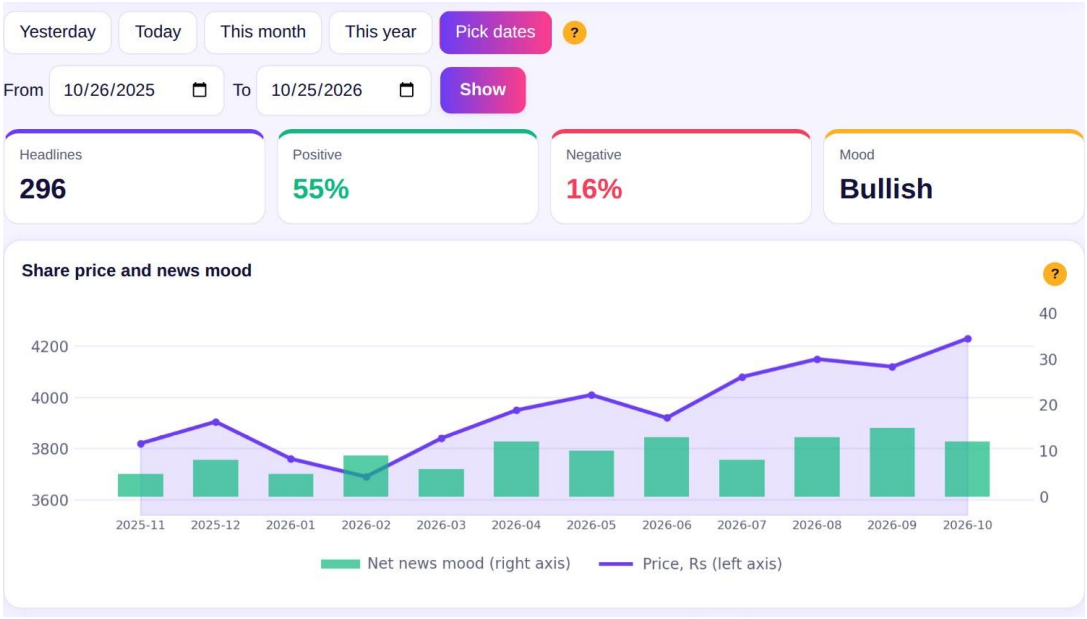


Figure 4.5: Timeline Selection, Summary Cards and Price vs News Mood Chart

The stacked bar chart shows daily or monthly counts of positive, neutral and negative headlines, the doughnut chart shows the share of each mood, and the horizontal bar chart shows the newspapers that published the most headlines.



Figure 4.6: Mood Charts and Top Newspapers

Every news card shows the newspaper logo and name, the date, a link to the article, the label with confidence, and the reason for the label with the signal words. A box above the list explains positive, neutral and negative.

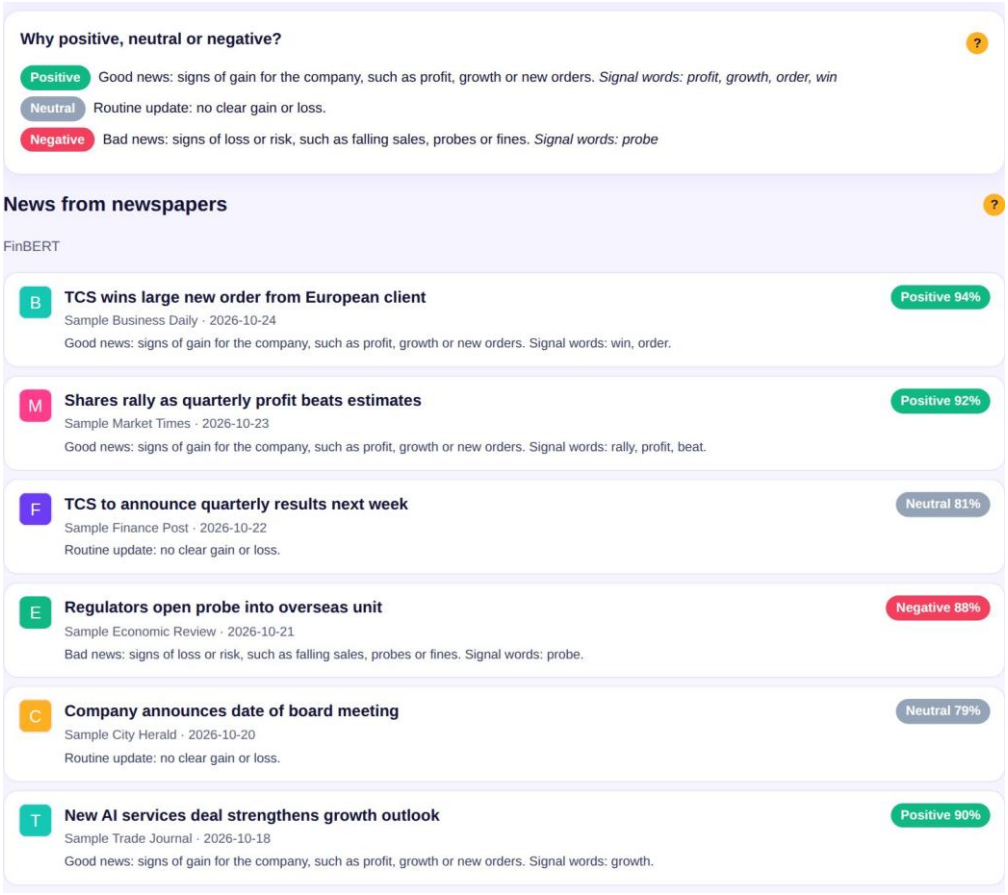


Figure 4.7: News Cards with Reasons for Each Label

When the user chooses Marathi, the font, the labels and the news edition change to Marathi.

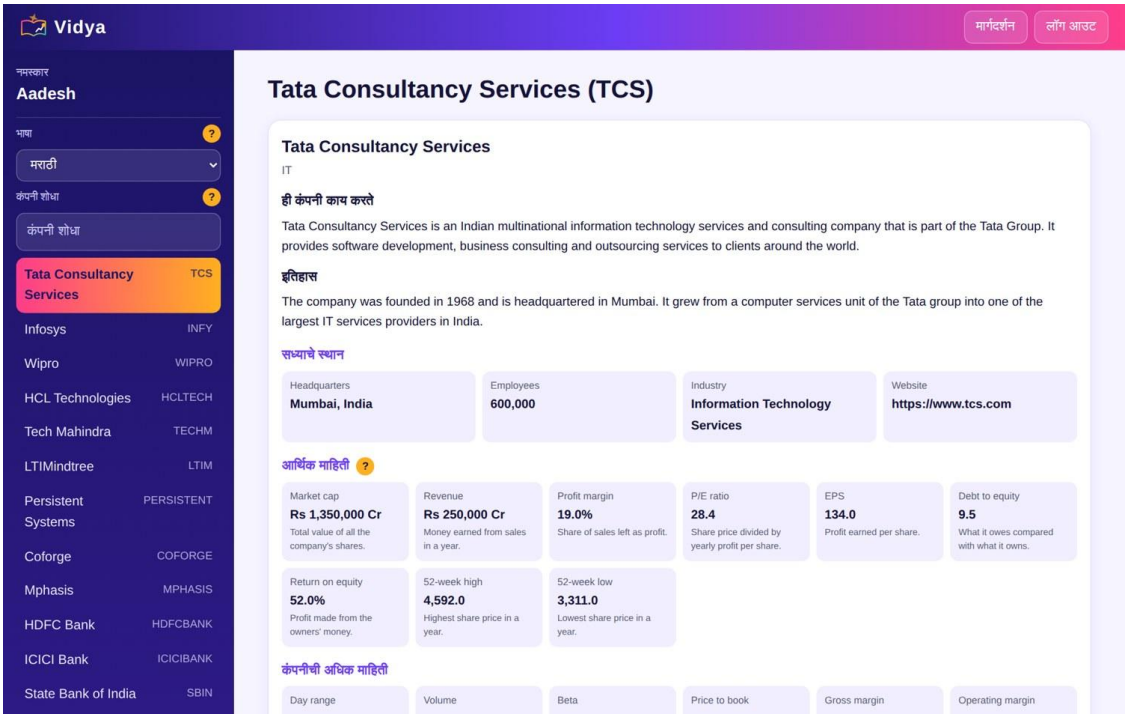


Figure 4.8: Marathi Interface



Figure 4.9: News in Marathi

The Guide button opens a step-by-step tour. The background is blurred and the section being explained is highlighted.

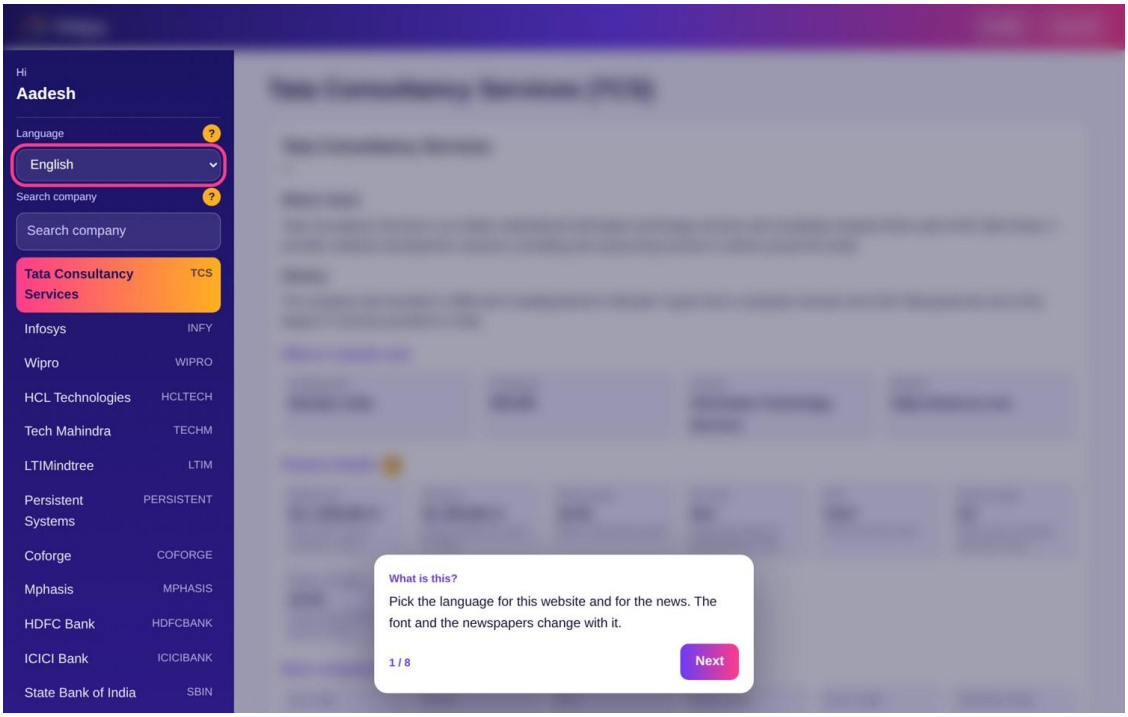


Figure 4.10: Guided Tour Pop-up with Blurred Background

Each small yellow question-mark button opens a pop-up that explains only that part, for example the price and mood chart.

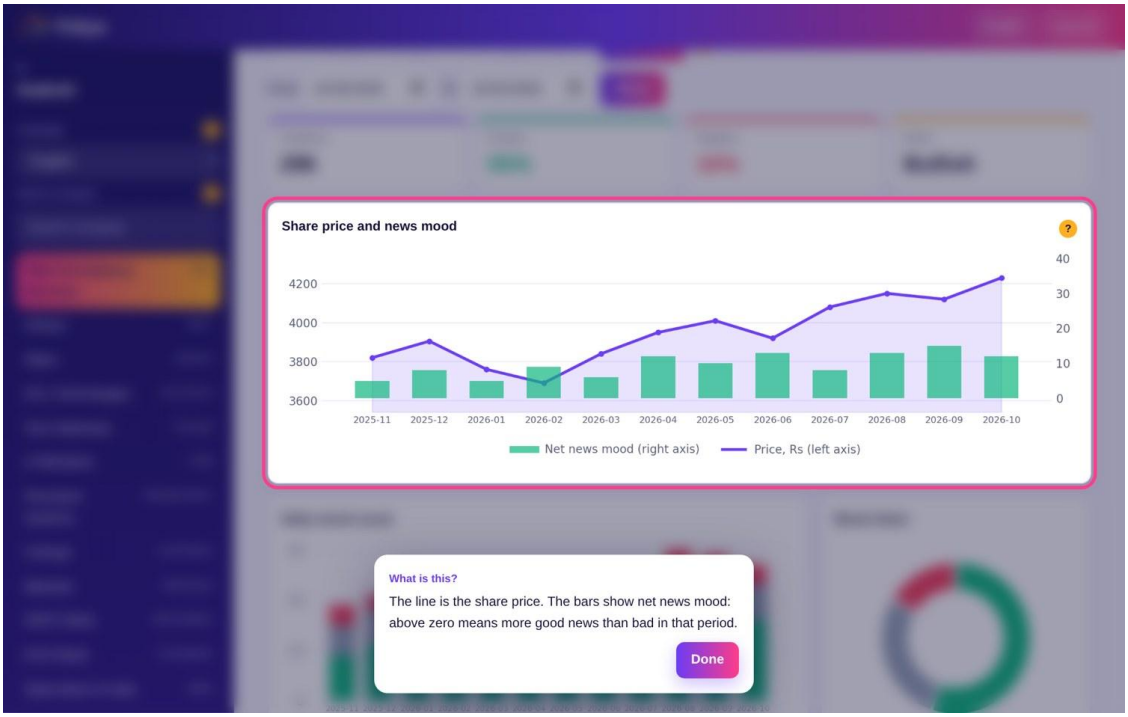


Figure 4.11: Help Pop-up for the Price and Mood Chart

4.2 Discussion: Comparative Study / Analysis

Table 4.1: Test Cases and Expected Results

No.	Test case	Input	Expected result
1	Language choice	Click Marathi on the first page	Font, labels and news edition change to Marathi
2	Register	Valid name, email and a password of 6 or more characters	Account is created and the dashboard opens
3	Weak password	Password with fewer than 6 characters	Error message is shown and no account is created
4	Wrong password	Registered email with a wrong password	Message that email or password is wrong
5	Guest entry	Click Continue as guest	Dashboard opens with the name Guest
6	Company search	Type bank in the sidebar search	List shows only matching companies
7	Company story	Click a company	Business, history, finance details and peers are shown
8	Today tab	Select Today	Headlines of the current date, or a no-news message
9	Custom dates	Start date in 2019, end date today	Monthly buckets and up to 100 headlines
10	Non-English news	Choose Marathi and a company	Marathi headlines with newspaper names, if the edition has coverage

No.	Test case	Input	Expected result
11	Label reason	Headline containing the word probe	Negative label with the reason and the signal word
12	Help pop-up	Click a question-mark button	Pop-up appears, background is blurred and section is highlighted
13	No internet	Disconnect the network	Friendly message is shown and the page does not crash

Table 4.2: Comparative Analysis

Feature	Manual reading	English-only sentiment tool	Vidya
News language	Reader's own	English	Twelve Indian languages
Reason for label	Reader's judgement	Score only	Plain reason and signal words
Company background	Separate search	Not provided	Story, finance and peers on one page
Older news	Archive search	Limited	Any date range
Newspaper source	By paper	Often missing	Name, logo and link
Cost	Time	Often paid	Free data sources

The results show that the system meets its objectives. Headlines are collected for the chosen company, language and dates, labelled by a language model, explained, and presented as graphs together with company information. The pop-up help and the language-first start page make the website easier to use for a first-time user.

Limitations: FinBERT is trained on English financial text, so the labels for headlines in other languages come from a general multilingual model and may be less accurate. Google News returns up to 100 headlines per search, so a very wide date range shows a sample and not every article. The company story is matched from Wikipedia by name and should be checked through the source link. Yahoo Finance data comes through an unofficial library and may sometimes be unavailable. The interface text is translated into three languages only, and the website is intended for learning and not for investment advice.

CHAPTER 5
Conclusion and Future Work

5.1 Conclusion

This project developed Vidya, a multilingual financial news sentiment website for 100 listed Indian companies. News headlines are collected in the language chosen by the user, classified as positive, neutral or negative by FinBERT or by a multilingual model, and explained in plain words. The website combines this with the company's business story, history, financial ratios and a ranked comparison with world peers, and shows the results in interactive graphs with a flexible date timeline. A language-first start page, registration, login and guest access, and pop-up help with a blurred background make the system friendly for beginners.

The work shows that modern language models can be put into a simple web application that helps ordinary readers study how the news is treating a company, in their own language and with reasons. The project also gave practical experience with NLP, REST APIs, databases, data visualisation and web design.

5.2 Future Scope

- Fine-tuning a multilingual model on labelled Indian-language financial news to improve the labels outside English.
- Translating the full interface into all twelve supported languages.
- Reading article images and summaries so that every news card can show a picture.
- Adding a personal watchlist for each registered user and alerts when the mood of a company changes strongly.
- Storing the scored results on a tamper-proof blockchain ledger shared by several computers, so that no single computer controls the data.
- Deploying the system on a cloud server and adding more companies, sectors and international markets.

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